Experiment 1. **Reactions of Acids**

1. Carry out the following reactions
2. Record your observations for each reaction
3. Name the products formed for each reaction
4. Write an ionic equation for each reaction

Solid copper carbonate + hydrochloric acid solution

Observations:

Products:

Ionic equation:

Solid sodium hydrogencarbonate + nitric acid solution

Observations:

Products:

Ionic equation:

Zinc + hydrochloric acid solution

Observations:

Products:

Ionic equation:

Solid iron sulfide + sulfuric acid solution

Observations:

Products:

Ionic equation:

Solid sodium sulfite + hydrochloric acid solution

Observations:

Products:

Ionic equation:

Reactions of Acids - Problem Sheet.

1. Complete the following equations by supplying the formulae of the missing substances:

   a. hydrochloric acid + zinc oxide → .....…… + salt solution

   b. nitric acid + potassium hydrogencarbonate → .....……. + ...…… + salt solution

   c. hydrochloric acid + magnesium → …… + salt solution

   d. sodium sulfite + sulfuric acid → .....……. + ...…… + salt solution

   e. nitric acid + sodium sulfide → .....…….. + salt solution

   f. magnesium carbonate + nitric acid → .....……. + ...…… + salt solution

   g. sulfuric acid + potassium sulfite → .....……. + ...…… + salt solution

   h. hydrochloric acid + magnesium hydrogencarbonate → .....……. + ...…… + salt solution

2. Give the products (names or formulae) that are formed in the following reactions:

   a. hydrochloric acid + potassium hydroxide → ..... + solution of .....

   b. calcium sulfite + nitric acid → .....……. + ...…… + solution of .....

   c. sulfuric acid + magnesium sulfide → .....…….. + solution of .....

   d. sodium hydrogencarbonate + nitric acid → .....… + ...…… + solution of .....

   e. hydrochloric acid + sodium sulfite → ..... + ...…… + solution of .....

   f. hydrochloric acid + magnesium carbonate → .....……. + ...…… + solution of .....

   g. nitric acid + magnesium oxide → ..... + solution of .....

   h. sulfuric acid + potassium sulfide→ ..... + solution of .....

   i. hydrochloric acid + barium → ..... + solution of .....

3. Write an ionic equation for each of the following reactions:

   a. hydrochloric acid solution is added to solid magnesium oxide

   b. dilute nitric acid solution is added to a solution of potassium sulfite

   c. sulfuric acid solution is added to a solution of sodium hydrgencarbonate

   d. hydrochloric acid solution is added to solid copper sulfide

3. (cont) e. dilute hydrochloric acid is added to magnesium

f. dilute sulfuric acid is added to potassium hydroxide solution

g. sulfuric acid solution is added to sodium sulfide solid

h. dilute sulfuric acid is added to potassium

i. solid magnesium sulfite is added to dilute hydrochloric acid

j. nitric acid solution is added to potassium sulfide solution

4. Describe what you would expect to see happening during the reactions

a. dilute hydrochloric acid solution is added to a solution of sodium hydrogencarbonate

b. sulfuric acid solution is added to solid copper sulfide

c. nitric acid solution is added to some magnesium sulfite

Answers:

1. a) H_2O
   b) $CO_2 + H_2O$
   c) H_2
   d) $SO_2 + H_2O$
   e) H_2S
   f) $CO_2 + H_2O$
   g) $SO_2 + H_2O$
   h) $CO_2 + H_2O$

2. a) water + solution of potassium chloride
   b) sulfur dioxide + water + solution of calcium nitrate
   c) hydrogen sulfide + solution of magnesium sulfate
   d) carbon dioxide + water + solution of sodium nitrate
   e) sulfur dioxide + water + solution of sodium chloride
   f) carbon dioxide + water + solution of magnesium chloride
   g) water + solution of magnesium nitrate
   h) hydrogen sulfide + solution of potassium sulfate
   i) hydrogen + solution of barium chloride

3. a) $2H^+ + MgO \rightarrow H_2O + Mg^{2+}$
   b) $2H^+ + SO_3^{2-} \rightarrow SO_2 + H_2O$
   c) $H^+ + HCO_3^- \rightarrow CO_2 + H_2O$
   d) $2H^+ + CuS \rightarrow H_2S + Cu^{2+}$
   e) $2H^+ + Mg \rightarrow H_2 + Mg^{2+}$
   f) $H^+ + OH^- \rightarrow H_2O$
   g) $2H^+ + Na_2S \rightarrow H_2S + 2Na^+$
   h) $2H^+ + 2K \rightarrow H_2 + 2K^+$
   i) $2H^+ + MgSO_3 \rightarrow SO_2 + H_2O + Mg^{2+}$
   j) $2H^+ + S^{2-} \rightarrow H_2S$

4. a) bubbles of colourless gas form
   b) rotten smelling gas produced, blue solution formed
   c) gas with irritating smell formed